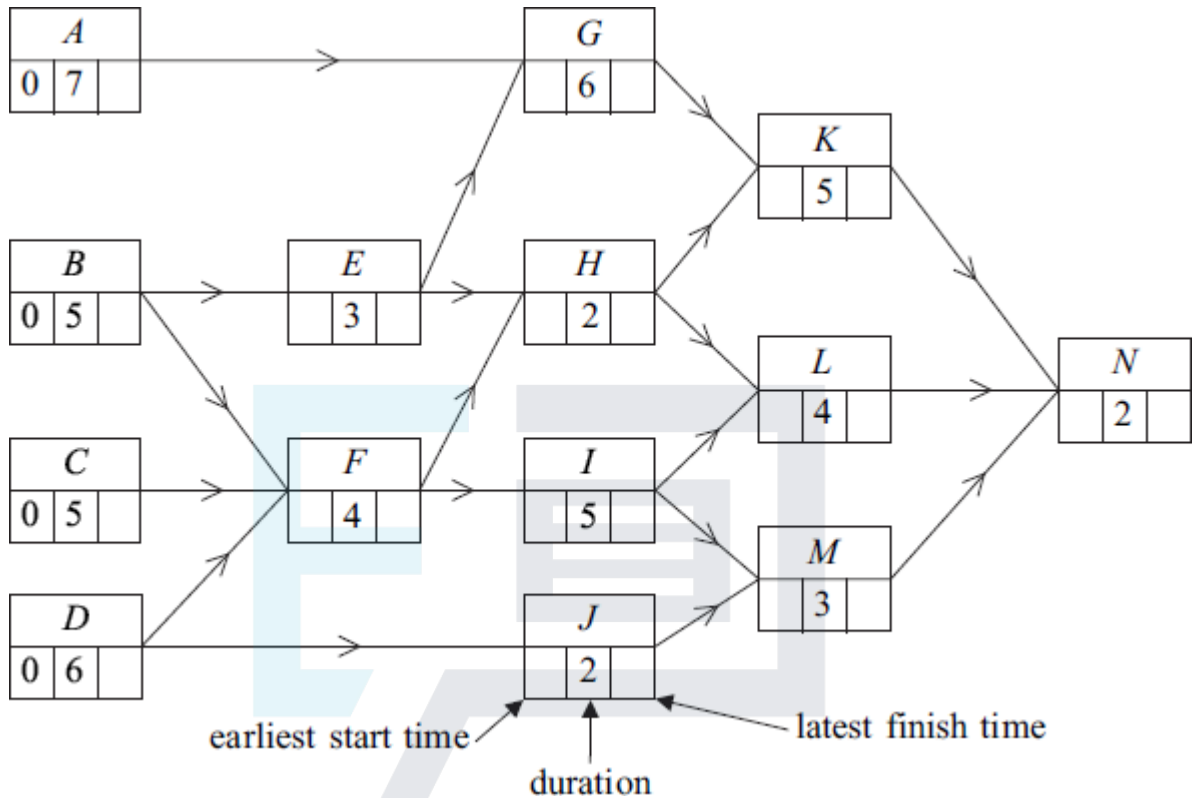


Q1.

Figure 1 below shows an activity diagram for a construction project. The time needed for each activity is given in days.

Figure 1



- (a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**. (4)

- (b) Find the critical paths and state the minimum time for completion of the project.

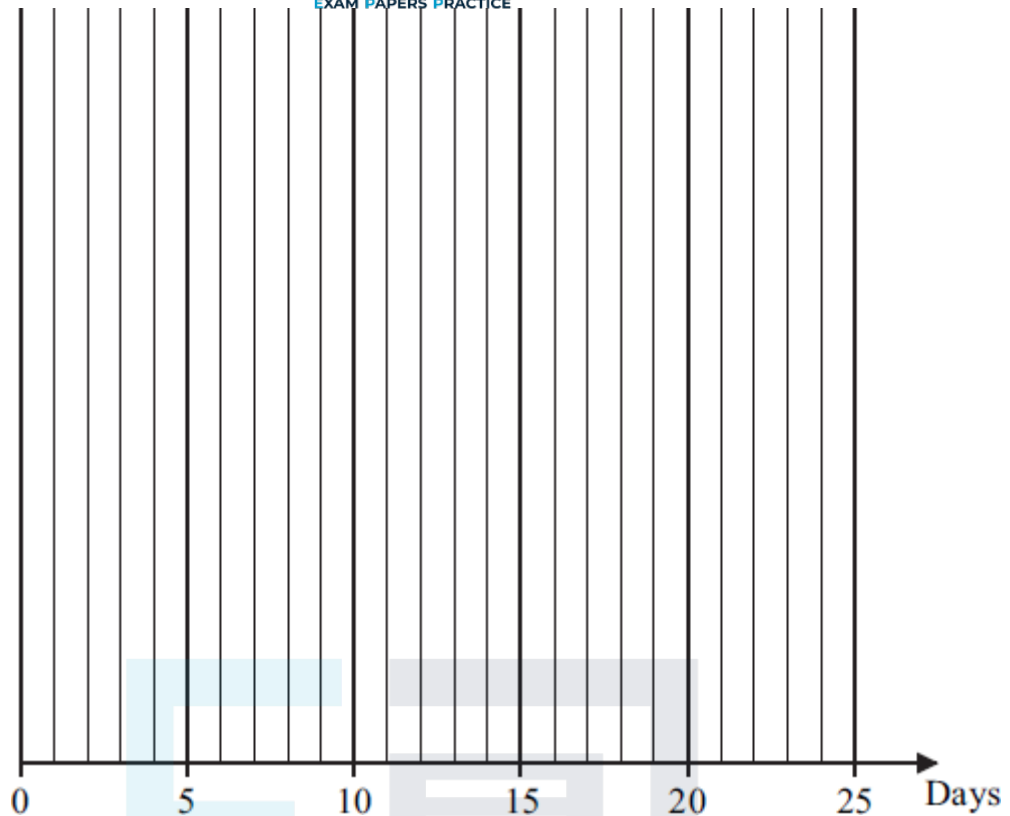
Critical paths are _____

Minimum completion time is _____ days.

(3)

- (c) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming that each activity starts as early as possible.

Figure 2



(5)

- (d) Activity J takes longer than expected so that its duration is x days, where $x \geq 3$. Given that the minimum time for completion of the project is unchanged, find a further inequality relating to the maximum value of x .

(2)

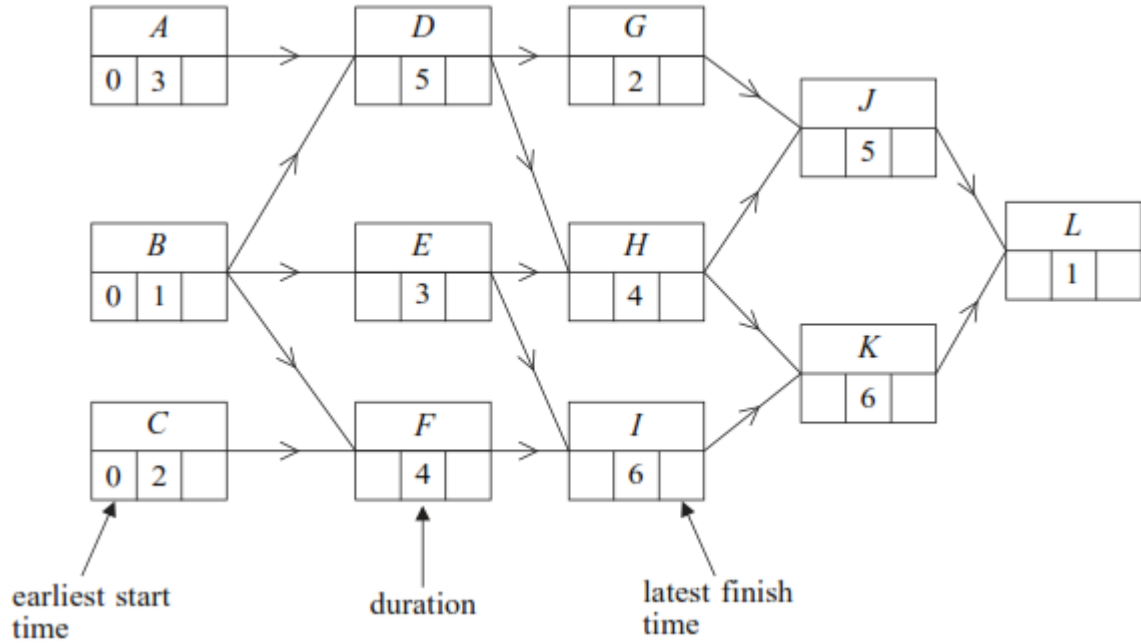
(Total 14 marks)

Q2.

Figure 1 below shows an activity diagram for a cleaning project. The duration of each activity is given in days.

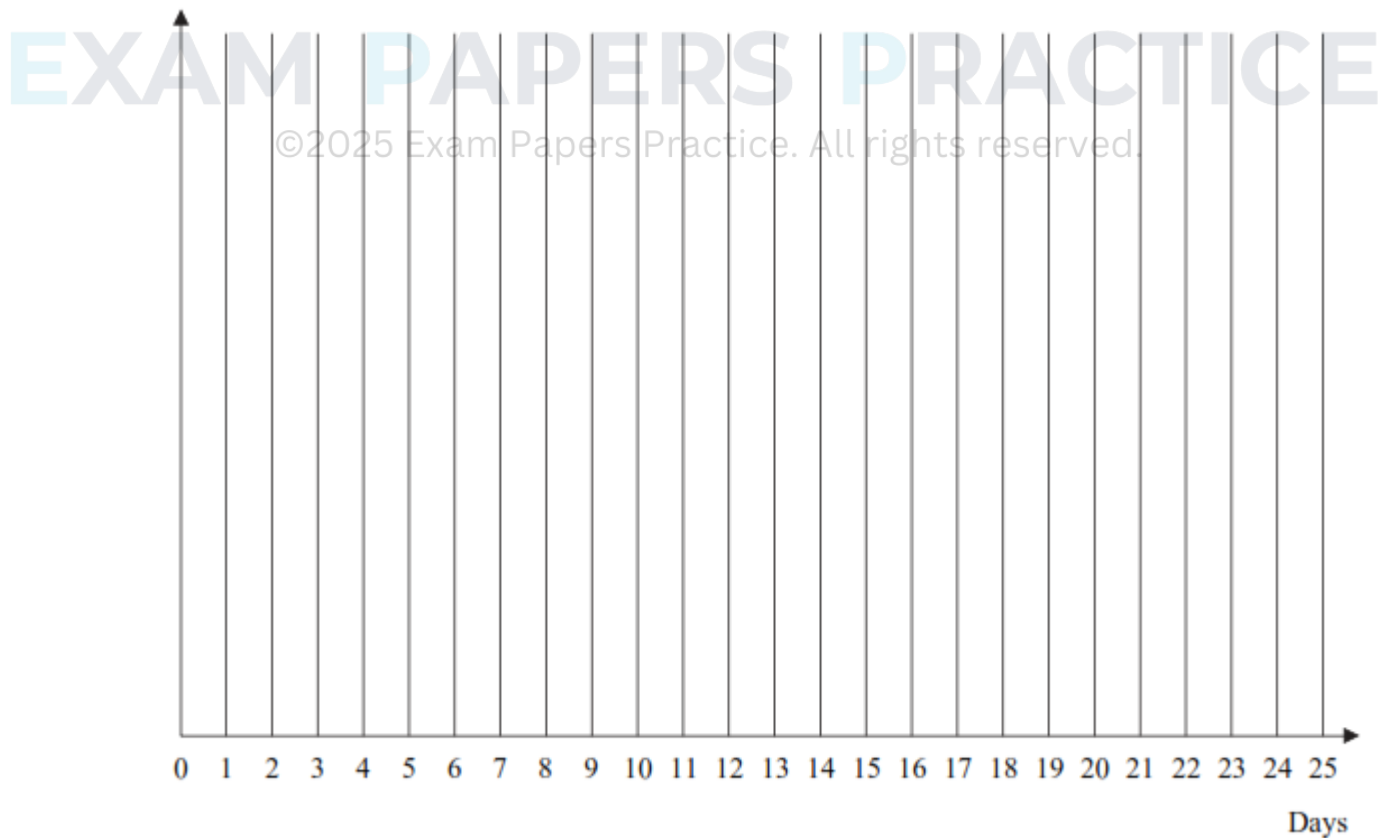
- (a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

Figure 1



- (4)
- (b) Find the critical paths and state the minimum time for completion of the project. (3)
- (c) Find the activity with the greatest float time and state the value of its float time. (2)
- (d) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming that each activity starts as **late** as possible.

Figure 2

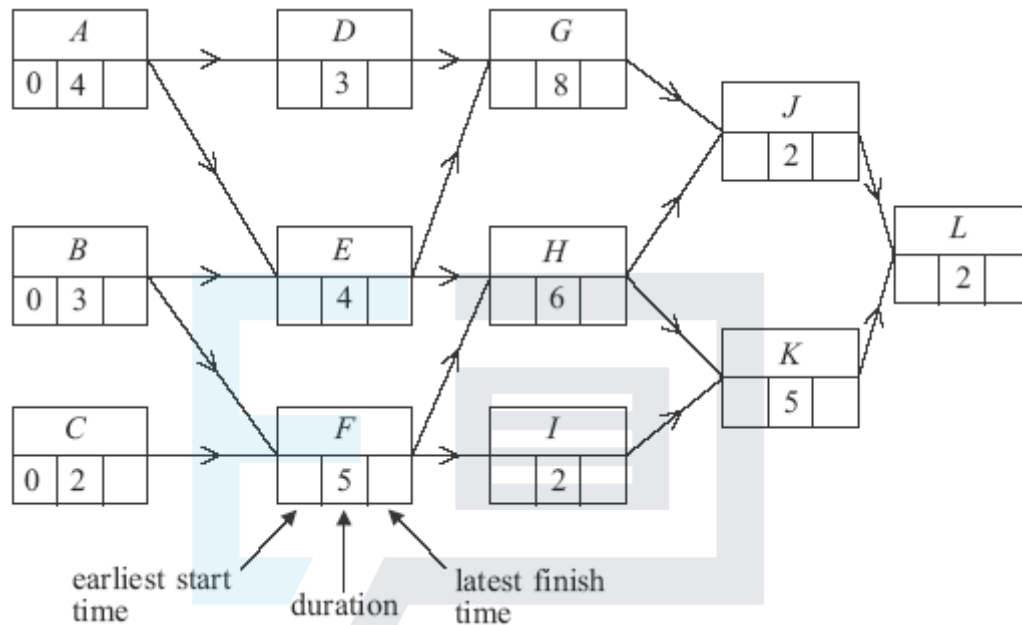


(4)
(Total 13 marks)

Q3.

Figure 1 below shows an activity diagram for a construction project. The time needed for each activity is given in days.

Figure 1



- (a) Find the earliest start time and latest finish time for each activity and insert their values on **Figure 1**.

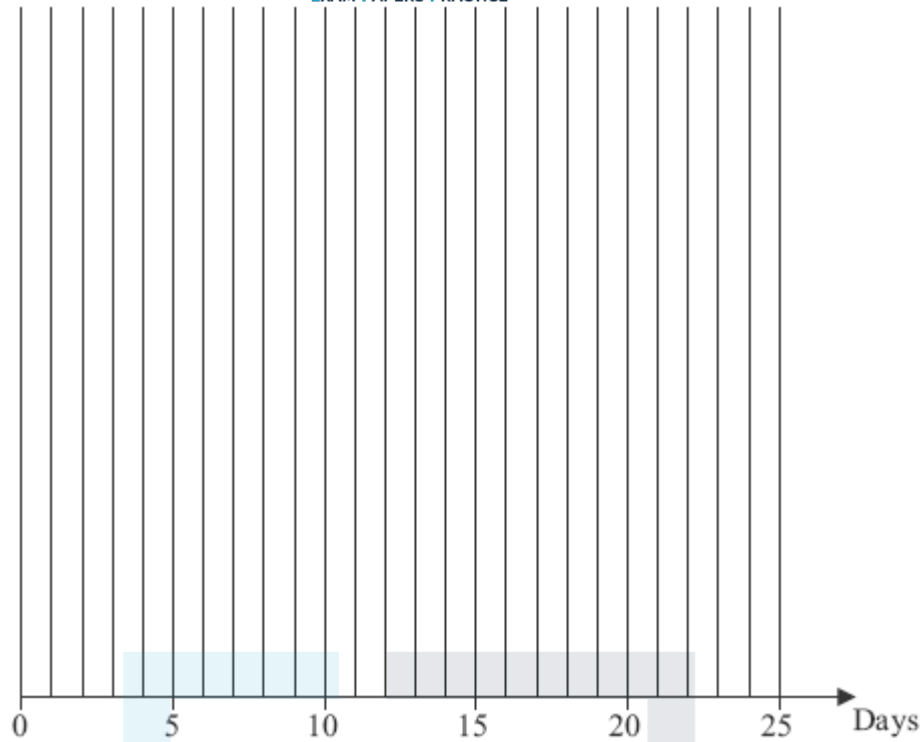
- (b) Find the critical paths and state the minimum time for completion of the project.

Critical paths are _____

Minimum completion time is _____ days

- (c) On **Figure 2** below, draw a cascade diagram (Gantt chart) for the project, assuming that each activity starts as early as possible.

Figure 2



(3)

(d) A delay in supplies means that Activity *I* takes 9 days instead of 2.

(i) Determine the effect on the **earliest** possible starting times for activities *K* and *L*.

(2)

(ii) State the number of days by which the completion of the project is now delayed.

(1)

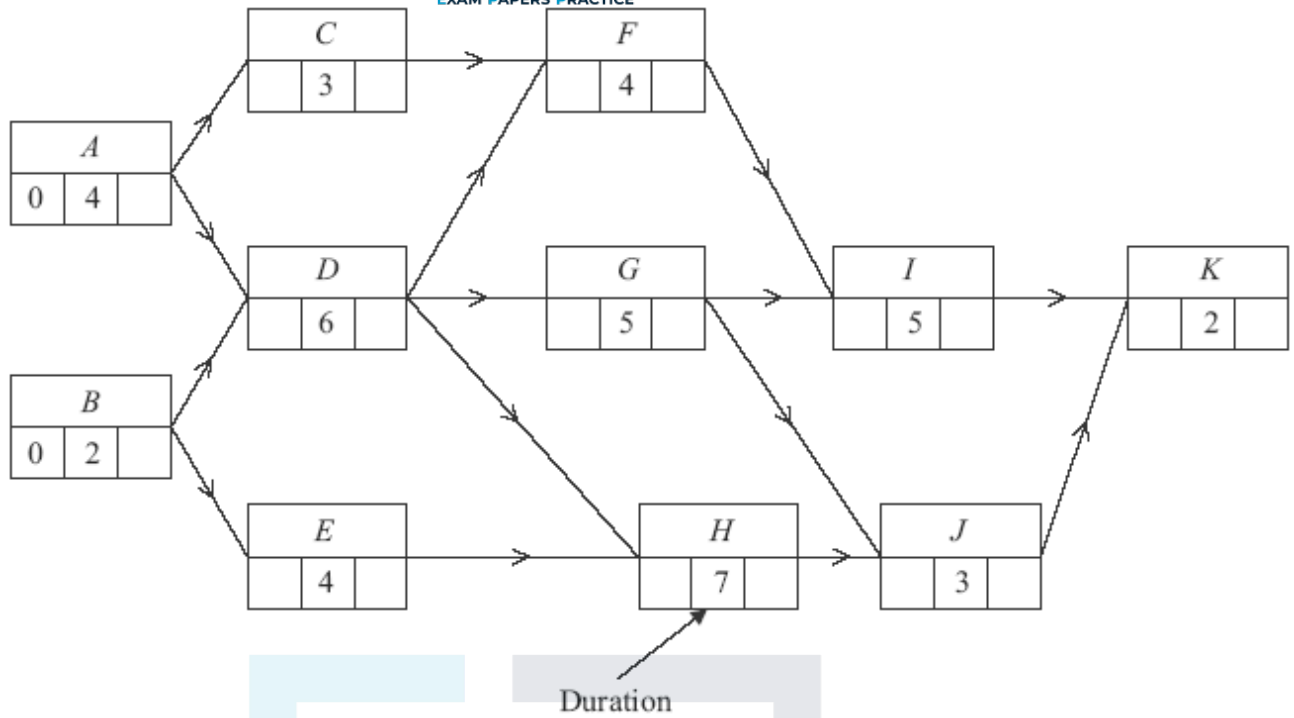
(Total 13 marks)

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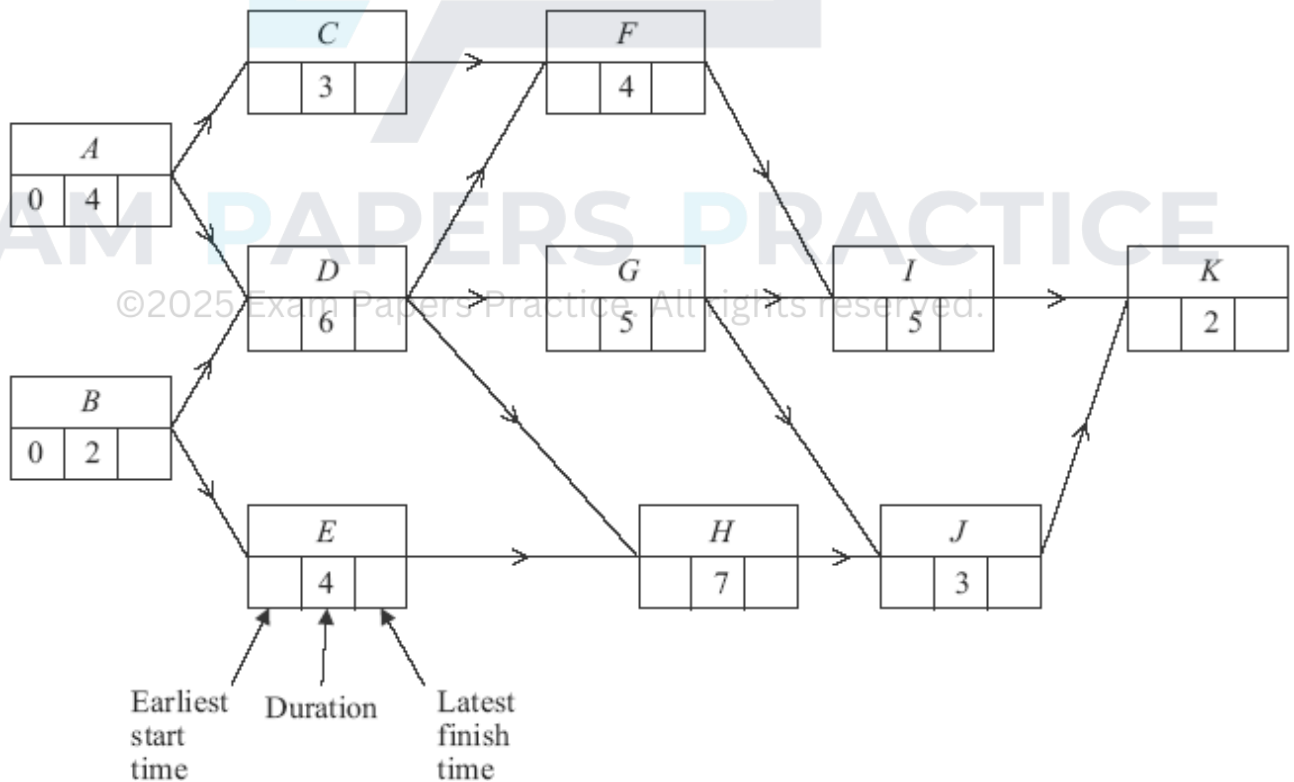
Q4.

The following diagram shows an activity network for a project. The time needed for each activity is given in days.



- (a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

Figure 1



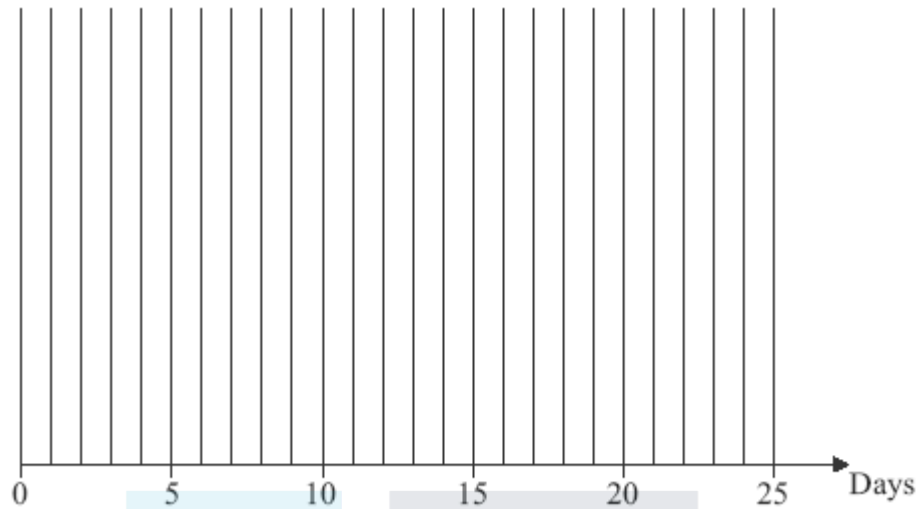
(4)

- (b) Find the critical paths and state the minimum time for completion.

(3)

- (c) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming each activity starts as early as possible.

Figure 2



(3)

- (d) Activity *C* takes 5 days longer than first expected. Determine the effect on the earliest start time for other activities and the minimum completion time for the project.

(2)

(Total 12 marks)

Q5.

[Figures 1 and 2, below, are provided for use in this question.]

Figure 1

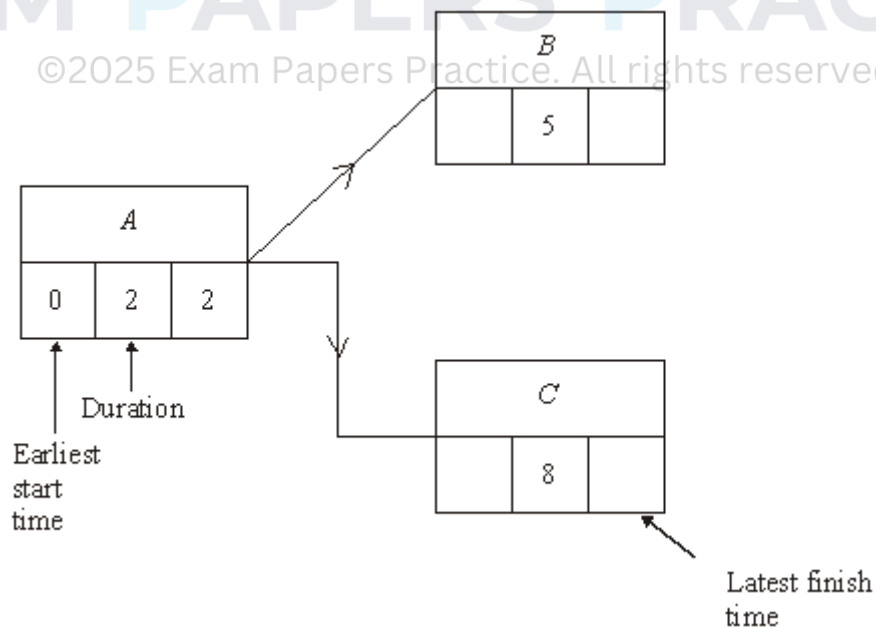
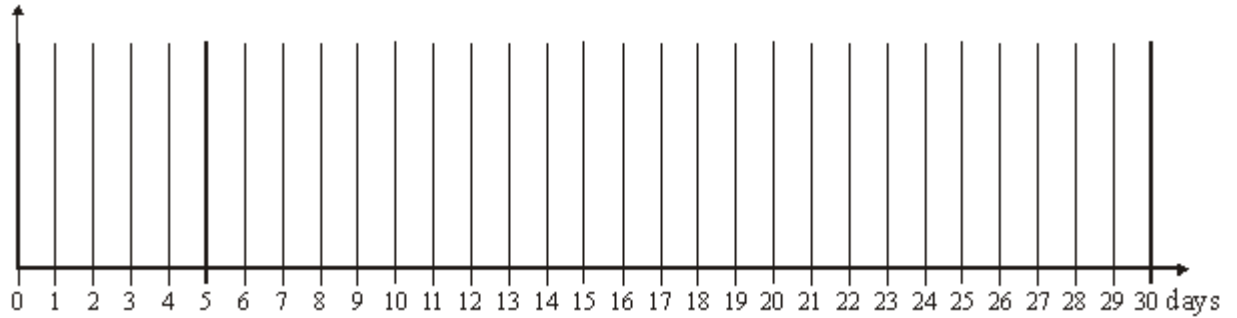


Figure 2



A construction project is to be undertaken. The table shows the activities involved.

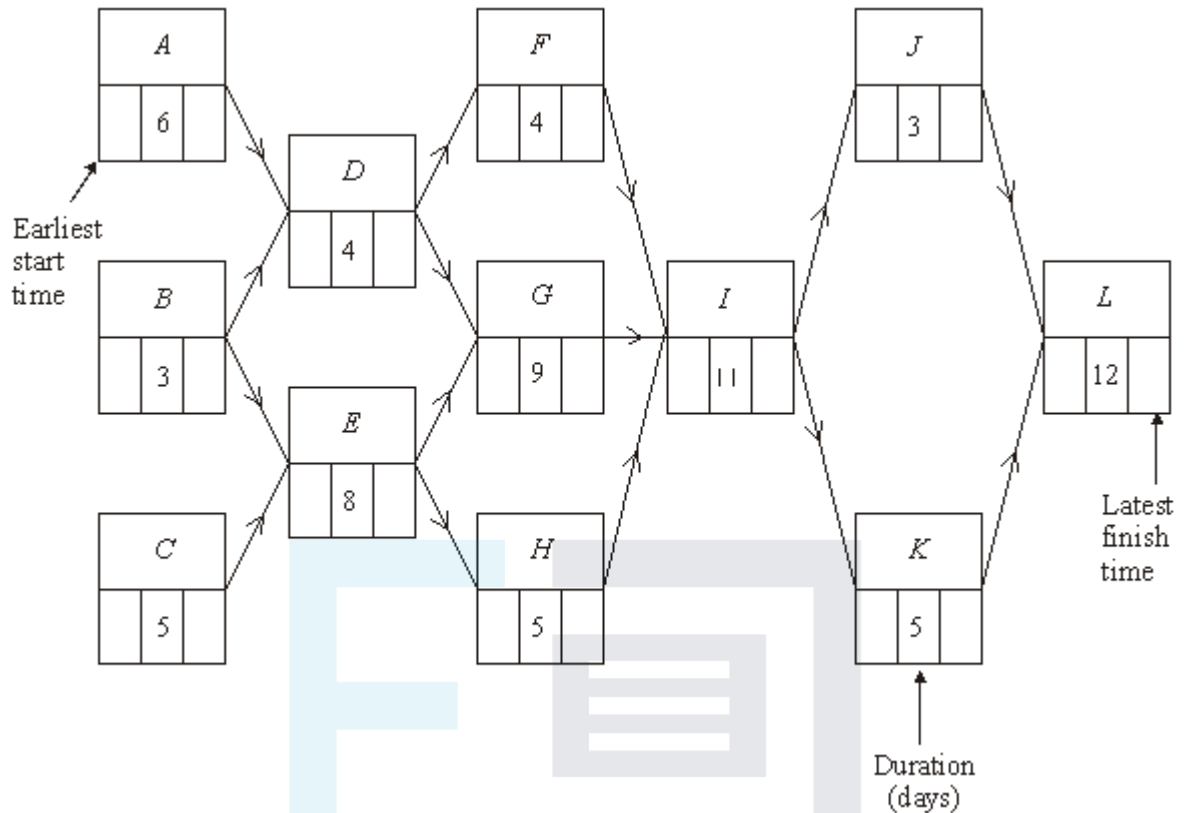
Activity	Immediate Predecessors	Duration (days)
<i>A</i>	–	2
<i>B</i>	<i>A</i>	5
<i>C</i>	<i>A</i>	8
<i>D</i>	<i>B</i>	8
<i>E</i>	<i>B</i>	10
<i>F</i>	<i>B</i>	4
<i>G</i>	<i>C, F</i>	7
<i>H</i>	<i>D, E</i>	4
<i>I</i>	<i>G, H</i>	3

- (a) Complete the activity network for the project on **Figure 1**. (3)
- (b) Find the earliest start time for each activity. (2)
- (c) Find the latest finish time for each activity. (2)
- (d) Find the critical path. (1)
- (e) State the float time for each non-critical activity. (2)
- (f) **On Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming each activity starts as **late** as possible. (4)

(Total 14 marks)

Q6.

The following diagram shows an activity network for a major building project.

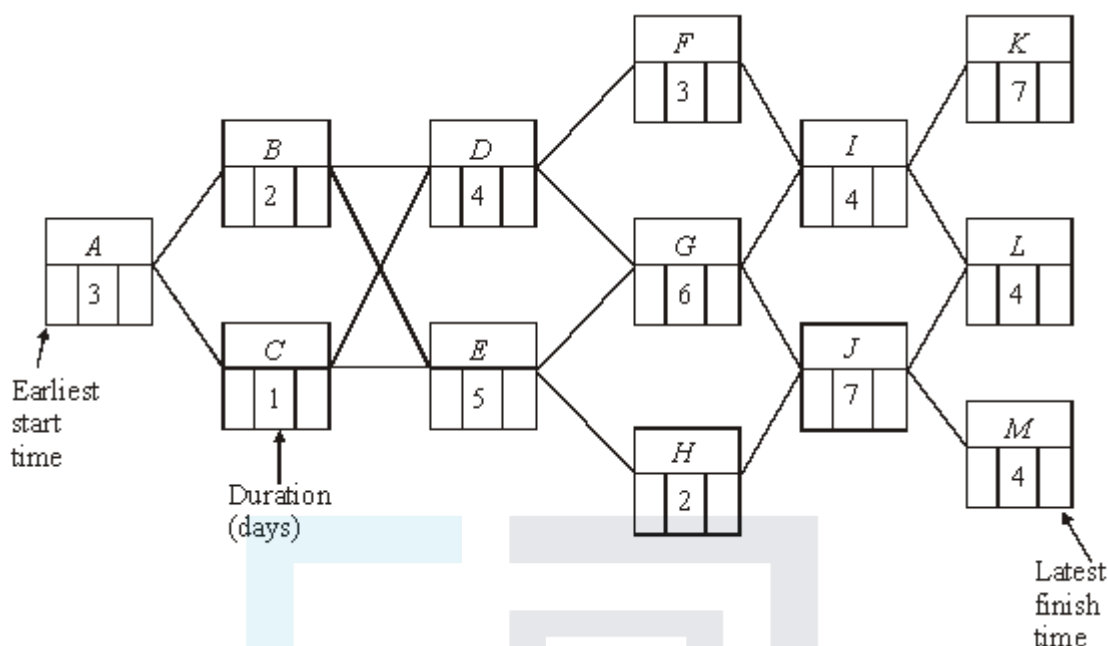


- (a) On the figure above find:
- the earliest start time for each activity; (2)
 - the latest finish time for each activity. (2)
- (b) Identify the critical path. (1)
- (c) State the activity with the greatest float time. (1)
- (d) Construct a Gantt (cascade) diagram for the project. (3)
- (e) Activities *D* and *J* overrun by 5 days each.
- Find:
- the new minimum completion time for the project; (3)
 - the new critical path. (1)

(Total 13 marks)

Q7.

A major construction project is to be undertaken and the project has been divided into 13 activities. The following diagram shows the activity network for the project.



- (a) On the diagram above:
- find the earliest start time for each activity;
 - find the latest finish time for each activity.
- (b) List the non-critical activities.
- (c) Draw a Gantt (cascade) diagram for the project.
- (d) There are only two workers available, each able to complete any of the activities in the stated time.
- Find the new minimum completion time for the whole project.
 - Allocate the activities to the two workers so that this new minimum time can be achieved.

(Total 12 marks)

Q8.

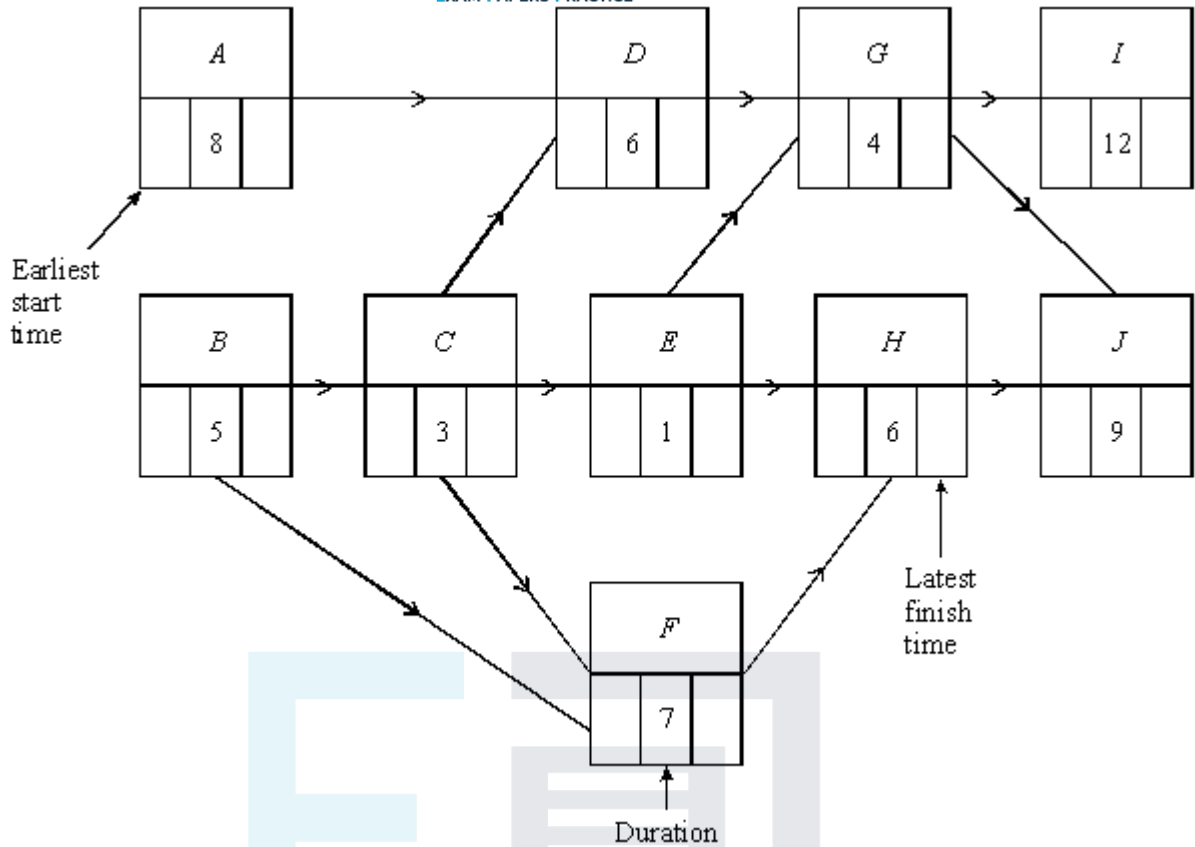
A building company is going to lay a new block-paved driveway. The table below shows the activities involved.

	Activity	Immediate Predecessor	Duration (days)
<i>A</i>	Remove tarmac top drive	—	7
<i>B</i>	Remove tarmac bottom drive	—	6
<i>C</i>	Concrete edging top	<i>A</i>	3
<i>D</i>	Concrete edging bottom	<i>B</i>	3
<i>E</i>	Lay hardcore	<i>C, D</i>	2
<i>F</i>	Lay sand	<i>E</i>	2
<i>G</i>	Lay blocks	<i>F</i>	5
<i>H</i>	Brush sand	<i>G</i>	1

- (a) Construct an activity network for the project. (3)
- (b) Find the minimum completion time for the whole project. (2)
- (c) Given that each activity starts as early as possible, draw a cascade (Gantt) chart for the project. (4)
- (Total 9 marks)

Q9.

The following diagram shows an activity network for a project. The time required for each activity is given in hours.



(a) On the table below, complete the precedence table.

Activity	Immediate Predecessor
A	
B	
C	
D	
E	
F	
G	
H	
I	
J	

(2)

(b) On the diagram above, complete the earliest start and latest finish times for each activity.

(4)

(c) List the critical activities.

(1)

(d) Draw a Gantt diagram, given that the project is to be completed in the minimum possible time and all the activities are to start as late as possible.

(4)

(c) There are only two workers available. They are both capable of completing each activity on their own in the stated time.

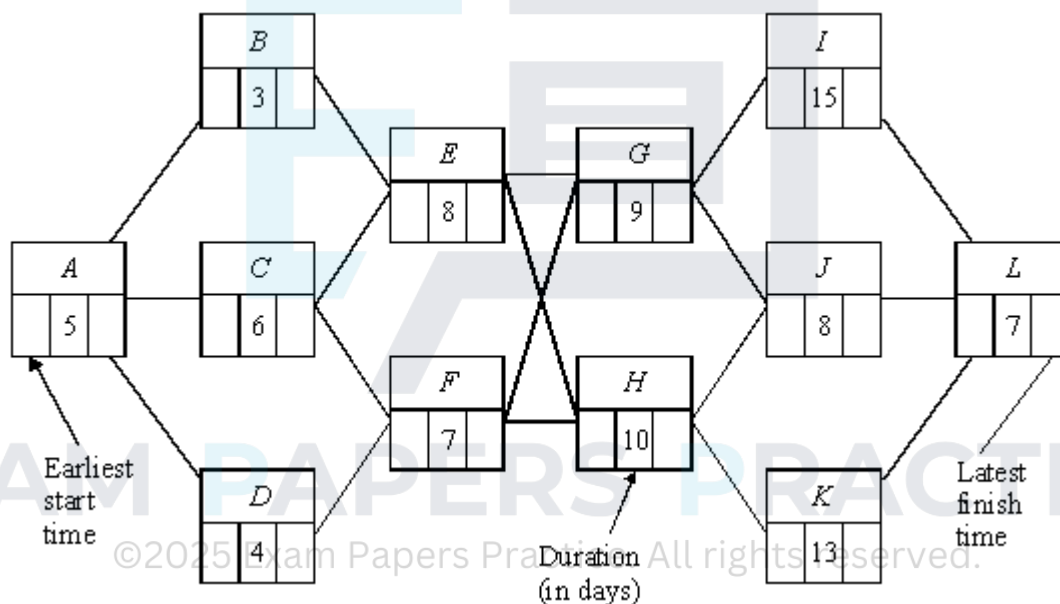
Find the minimum overrun time for the whole project.

(2)

(Total 13 marks)

Q10.

The following diagram shows an activity network for a major building programme.



(a) On diagram,

(i) find the earliest start time for each activity,

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Identify the critical path.

(1)

(c) Construct a Gantt diagram for the project.

(3)

(d) Given that at most two activities can take place at the same time, find the new minimum completion time for the whole project

(3)

Q11.

The shell of a flat has been constructed and the inside is to be completed. The work has been divided into 13 activities, as shown in the table below.

Activity		Tradesman	Immediate predecessors	Duration (days)
<i>A</i>	Studding	Builder		2
<i>B</i>	Electrics (first fix)	Electrician	<i>A</i>	1
<i>C</i>	Plumbing (first fix)	Plumber	<i>A</i>	1.5
<i>D</i>	Plaster boarding	Plasterer	<i>B, C</i>	1
<i>E</i>	Plastering	Plasterer	<i>D</i>	1
<i>F</i>	Artexing	Decorator	<i>E</i>	1
<i>G</i>	Electrics (second fix)	Electrician	<i>E</i>	0.5
<i>H</i>	Joinery (first fix)	Joiner	<i>E</i>	1
<i>I</i>	Plumbing (second fix)	Plumber	<i>E</i>	1
<i>J</i>	Joinery (second fix)	Joiner	<i>G, H, I</i>	2
<i>K</i>	Painting	Decorator	<i>F, J</i>	1
<i>L</i>	Tiling	Tiler	<i>I, J</i>	0.5
<i>M</i>	Cleaning	Cleaner	<i>K, L</i>	0.5

- (a) Construct an activity network for the project. (4)
- (b) Find the earliest start time for each activity. (3)
- (c) Find the latest finish time for each activity. (3)
- (d) Identify the critical activities and state the shortest completion time for the whole project. (2)
- (e) Construct a cascade (Gantt) diagram for the project, assuming that each activity is to start as early as possible. (3)

- (f) The foreman of the project can speed up the project by using either an extra electrician, or an extra plumber or an extra joiner.

An extra electrician would reduce the time for electrical work by 50%, an extra plumber would reduce the time for plumbing work by 25%, and an extra joiner would reduce the time for joinery work by 20%.

Explain which extra tradesman the foreman should use, and find the new minimum time to complete the project.

(5)
(Total 20 marks)

Q12.

[One sheet of 2 mm graph paper is provided for use in answering this question.]

Each year the secretary of a Mathematical Society undertakes the task of preparing and distributing the society's annual programme. The programme gives dates of meetings, a list of speakers and the titles of their talks, and an up-to-date list of paid-up members. The table below shows the activities which the secretary needs to carry out in preparing and distributing the programme.

	Activity	Immediate predecessor	Duration (days)
A	Select dates for meetings	—	3
B	Book speakers	A	13
C	Obtain advertising material for programme	A	12
D	Send out membership renewal notices	—	21
E	Prepare list of paid-up members	D	6
F	Prepare programme for printers and check proofs	B, C, E	10
G	Assemble material for circulation	F	7
H	Print members' address labels	E	4
I	Send out programmes	G, H	3

- (a) Construct an activity network for the project. (4)
- (b) On your network show full details of the earliest and latest starting and finishing times for each activity. Indicate and state the critical path. (6)
- (c) For each activity state the float time. (2)
- (d) Construct a cascade diagram for the project.
If each activity requires one helper so that activities are completed on time, state the minimum number of helpers required for the project. (5)
- (e) The secretary is let down by one helper who resigns immediately prior to the start of

the project and so cannot be replaced. Find the least possible number of days' delay to the completion of the project and explain how it can be achieved.

(2)

(Total 19 marks)

Q13.

The table shows the activities involved in a project, their durations, and their immediate predecessors.

Activity	A	B	C	D	E	F	G	H
Duration (days)	3	1	1	4	5	2	2	2
Immediate predecessors	–	–	B	A	A, C	B	F	D, E

(a) Draw an activity network for the project.

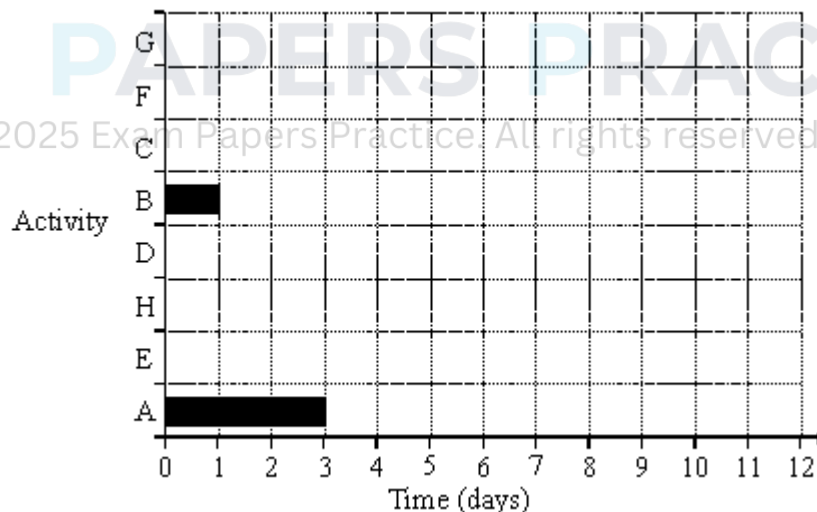
(5)

(b) Perform a forward pass and a backward pass on your activity network. Give the minimum completion time and the activities forming the critical path.

(6)

(c) In a cascade chart each activity is represented by a bar showing when it is scheduled. In the chart below, activities A and B are shown, both starting as early as possible.

Copy and complete the cascade chart, with the activities ordered as shown, and with each activity starting as early as possible.



(3)

(d) The number of people needed for each activity is as follows:

Activity	A	B	C	D	E	F	G	H
People	3	2	2	2	3	1	1	4

Activities are to be scheduled, some later than their earliest possible start time, so

that at most 5 people are needed at any one time, whilst the project is still completed in the minimum time.

What are the new scheduled start times for activities D, F and G?

(2)

(Total 16 marks)



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